VP02 — Architecture and logical decomposition

What parts exist and how are they organized?

Container/Unit and contains; library structure is not deployment allocation.

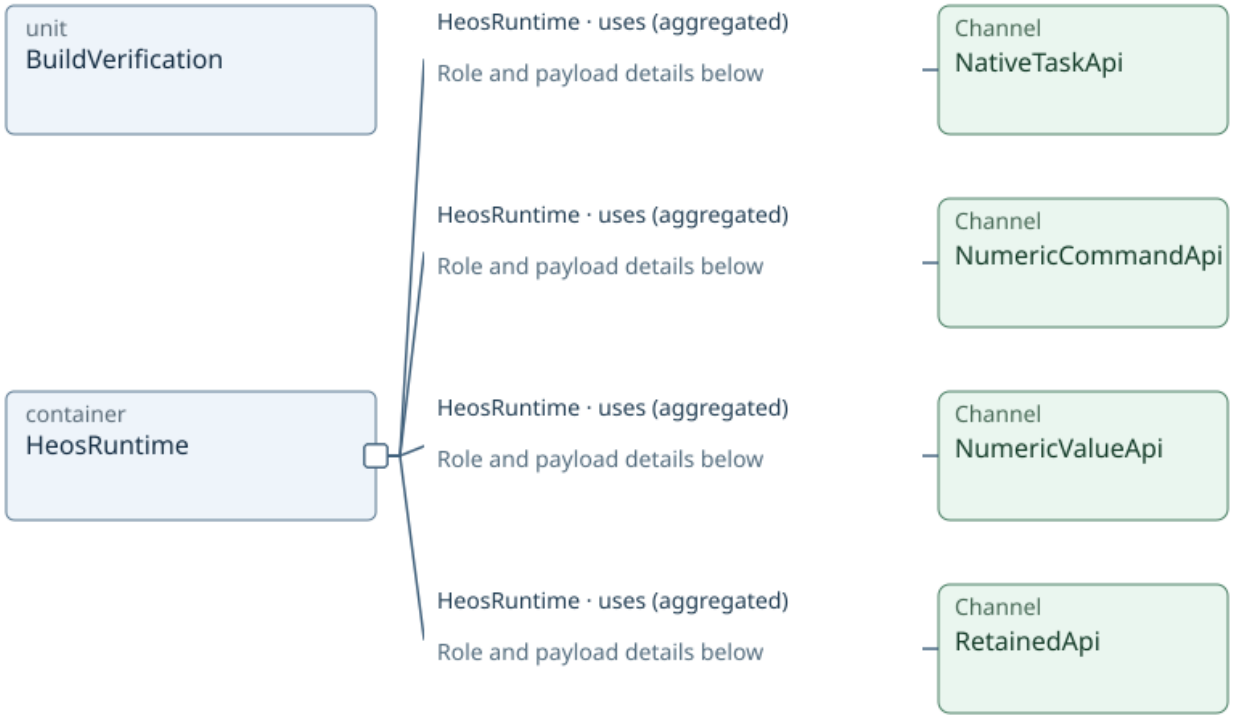
Operating mode: All declared modes · MicroConsole · MicroCore · MicroRetained

This page describes the checked model: 129 declarations and 384 source statements. Model claims do not establish implemented behavior or execution evidence.

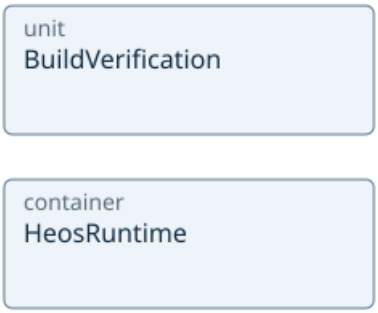
Structure and collaboration

System collaboration — aggregated

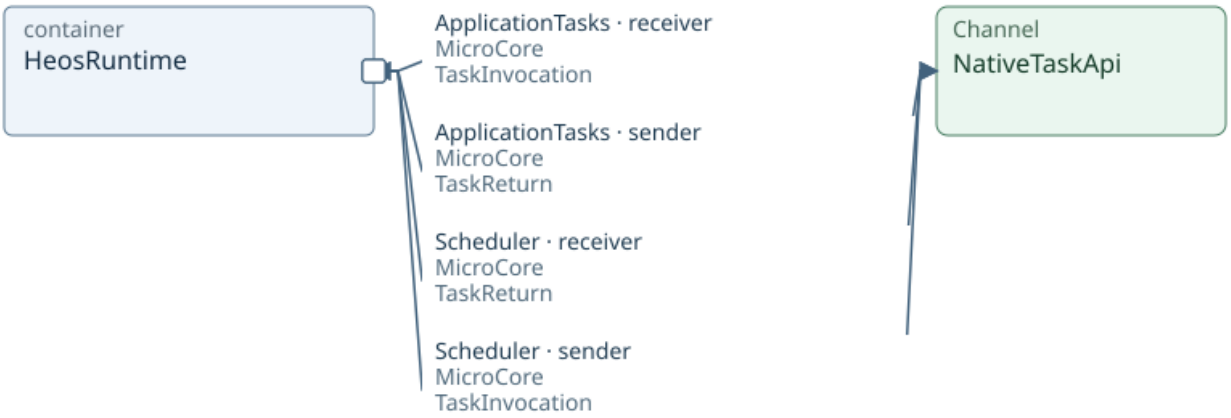
endpoints; roles and modes in the following figures



Containment — 1–2 of 2 objects



NativeTaskApi — mode MicroCore



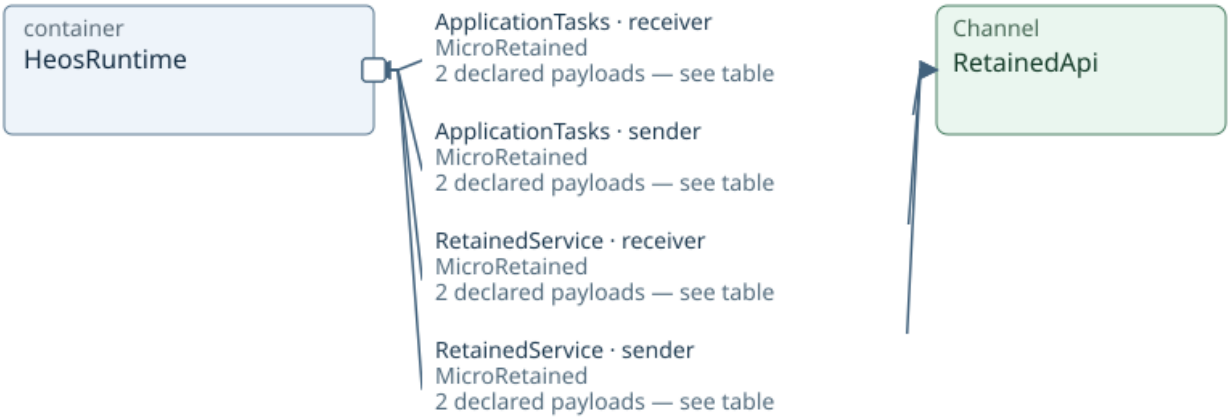
NumericCommandApi — mode MicroConsole



NumericValueApi — mode MicroConsole



RetainedApi — mode MicroRetained



Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: BuildVerification · HeosRuntime

NativeTaskApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	TaskInvocation	Not applicable	MicroCore
ApplicationTasks	sender	TaskReturn	Not applicable	MicroCore
Scheduler	receiver	TaskReturn	Not applicable	MicroCore
Scheduler	sender	TaskInvocation	Not applicable	MicroCore

NumericCommandApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
CommandRegistry	receiver	CommandRequest	Not applicable	MicroConsole
CommandRegistry	sender	CommandResult	Not applicable	MicroConsole
ShellTask	receiver	CommandResult	Not applicable	MicroConsole
ShellTask	sender	CommandRequest	Not applicable	MicroConsole

NumericValueApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ShellTask	receiver	ReadValueResult	Not applicable	MicroConsole
ShellTask	receiver	WriteValueResult	Not applicable	MicroConsole
ShellTask	sender	ReadValueRequest	Not applicable	MicroConsole
ShellTask	sender	WriteValueRequest	Not applicable	MicroConsole
ValueRegistry	receiver	ReadValueRequest	Not applicable	MicroConsole
ValueRegistry	receiver	WriteValueRequest	Not applicable	MicroConsole
ValueRegistry	sender	ReadValueResult	Not applicable	MicroConsole

ValueRegistry	sender	WriteValueResult	Not applicable	MicroConsole

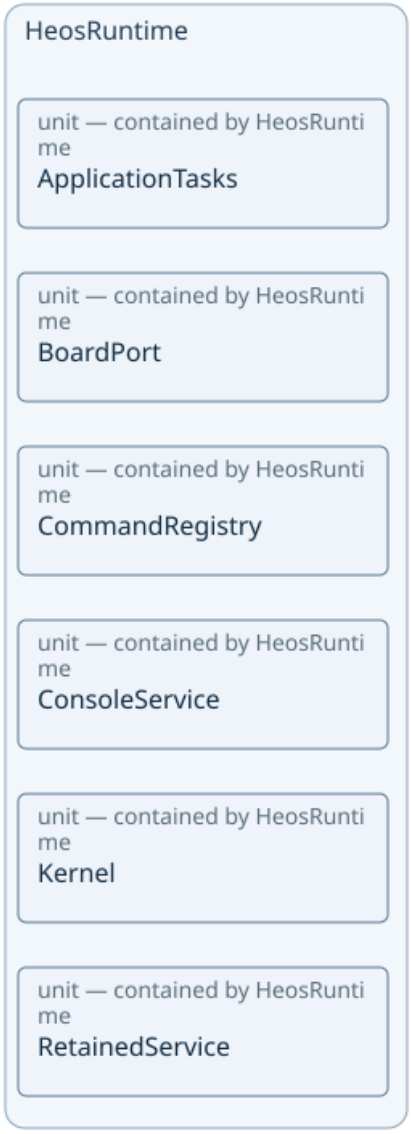
RetainedApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	CommitResult	Not applicable	MicroRetained
ApplicationTasks	receiver	RecoverResult	Not applicable	MicroRetained
ApplicationTasks	sender	CommitRequest	Not applicable	MicroRetained
ApplicationTasks	sender	RecoverRequest	Not applicable	MicroRetained
RetainedService	receiver	CommitRequest	Not applicable	MicroRetained
RetainedService	receiver	RecoverRequest	Not applicable	MicroRetained
RetainedService	sender	CommitResult	Not applicable	MicroRetained
RetainedService	sender	RecoverResult	Not applicable	MicroRetained

HeosRuntime — participants and

channels

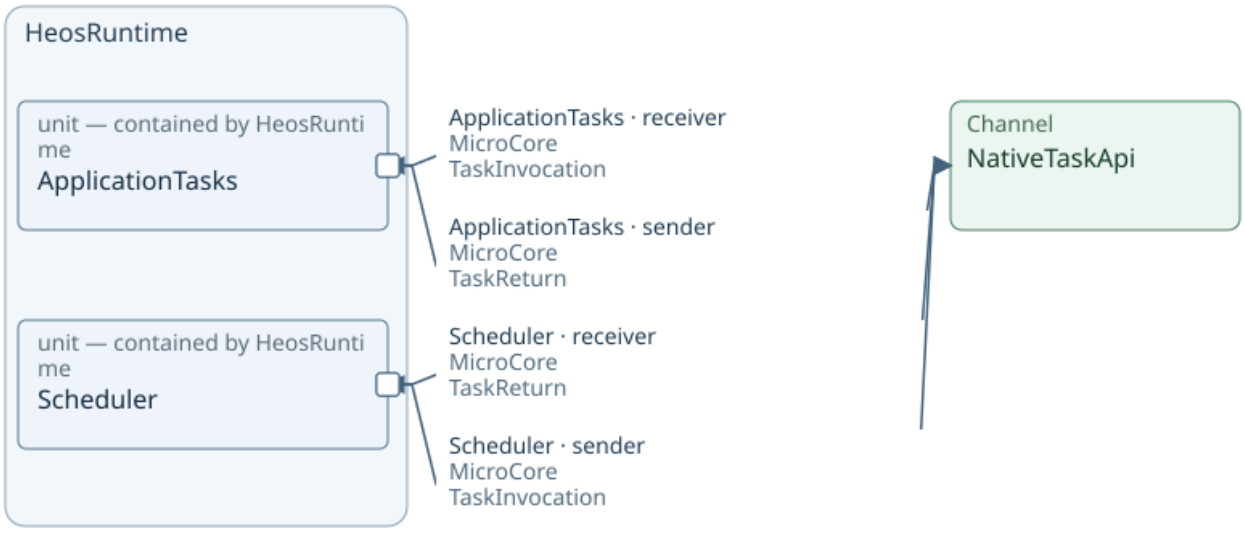
Containment — 1–6 of 9 objects



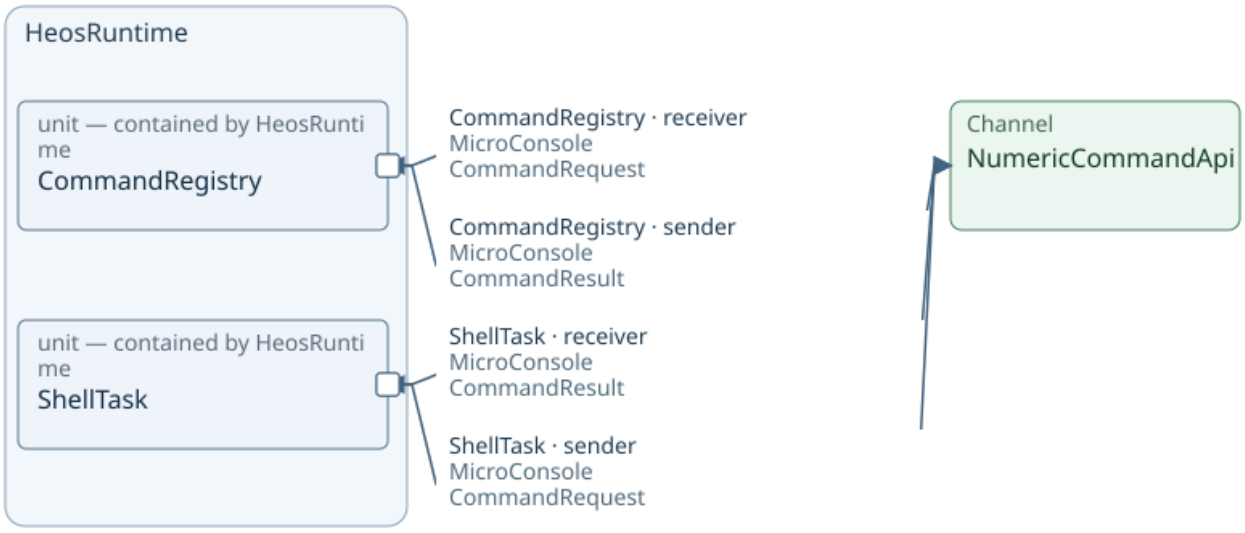
Containment — 7–9 of 9 objects



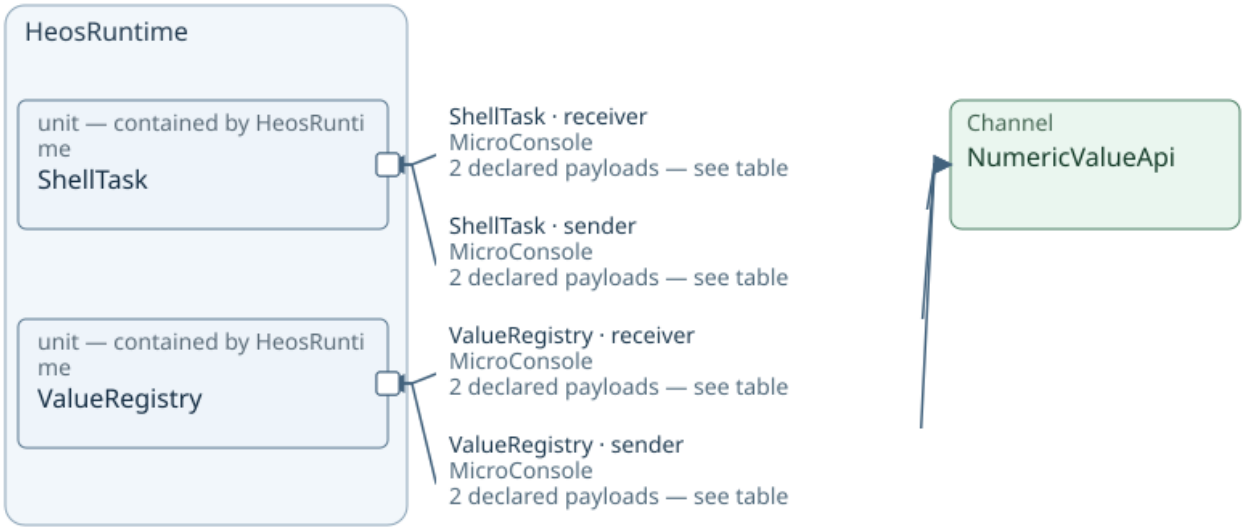
NativeTaskApi — mode MicroCore



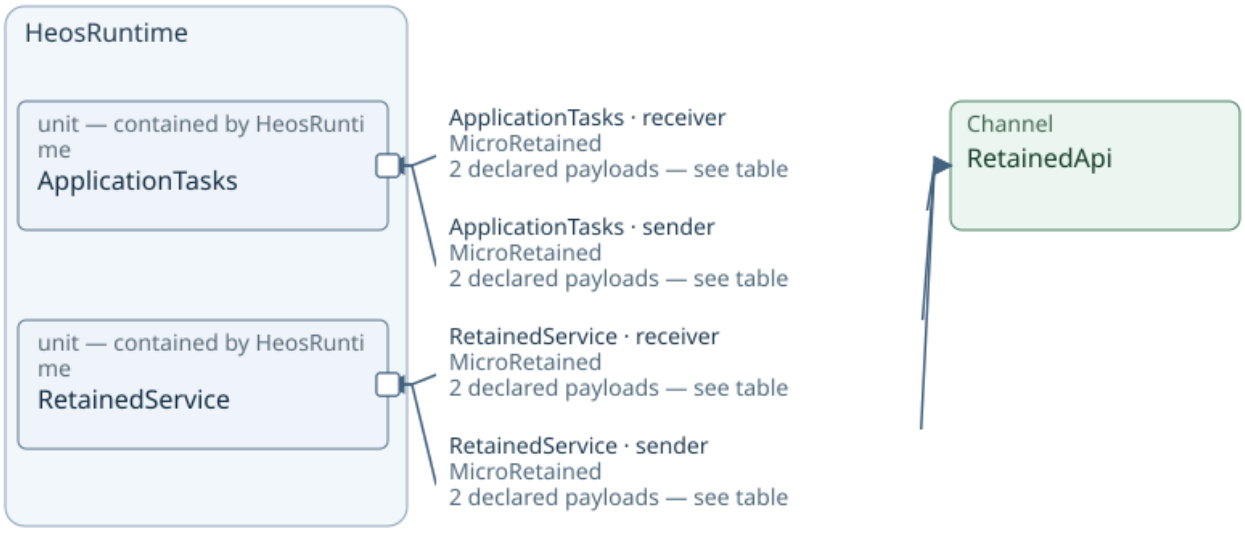
NumericCommandApi — mode MicroConsole



NumericValueApi — mode MicroConsole



RetainedApi — mode MicroRetained



Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: ApplicationTasks · BoardPort · CommandRegistry · ConsoleService · Kernel · RetainedService · Scheduler · ShellTask · ValueRegistry

NativeTaskApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	TaskInvocation	Not applicable	MicroCore
ApplicationTasks	sender	TaskReturn	Not applicable	MicroCore
Scheduler	receiver	TaskReturn	Not applicable	MicroCore
Scheduler	sender	TaskInvocation	Not applicable	MicroCore

NumericCommandApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
CommandRegistry	receiver	CommandRequest	Not applicable	MicroConsole
CommandRegistry	sender	CommandResult	Not applicable	MicroConsole
ShellTask	receiver	CommandResult	Not applicable	MicroConsole
ShellTask	sender	CommandRequest	Not applicable	MicroConsole

NumericValueApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ShellTask	receiver	ReadValueResult	Not applicable	MicroConsole
ShellTask	receiver	WriteValueResult	Not applicable	MicroConsole
ShellTask	sender	ReadValueRequest	Not applicable	MicroConsole
ShellTask	sender	WriteValueRequest	Not applicable	MicroConsole
ValueRegistry	receiver	ReadValueRequest	Not applicable	MicroConsole
ValueRegistry	receiver	WriteValueRequest	Not applicable	MicroConsole
ValueRegistry	sender	ReadValueResult	Not applicable	MicroConsole

ValueRegistry	sender	WriteValueResult	Not applicable	MicroConsole

RetainedApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	CommitResult	Not applicable	MicroRetained
ApplicationTasks	receiver	RecoverResult	Not applicable	MicroRetained
ApplicationTasks	sender	CommitRequest	Not applicable	MicroRetained
ApplicationTasks	sender	RecoverRequest	Not applicable	MicroRetained
RetainedService	receiver	CommitRequest	Not applicable	MicroRetained
RetainedService	receiver	RecoverRequest	Not applicable	MicroRetained
RetainedService	sender	CommitResult	Not applicable	MicroRetained
RetainedService	sender	RecoverResult	Not applicable	MicroRetained

Reading chapters

- Containers — 1 objects. HeosRuntime.
- Units — 10 objects. ApplicationTasks. BoardPort. BuildVerification.

Interpretation and unspecified behavior

Contains describes logical structure. Deployment, scheduling, runtime calls,

and unnamed internal units are not inferred. Channel endpoints refer to actual declared participants.